

SACH AI

Smart Assistant for Checking Honesty in Financial Information

"Suniye, Samajhiye, Sachai pe Bharosa Kijiye."

SANGYAN Hackathon • Track E (Misinformation & Content Literacy)

Organized by SNTC, IIT (BHU) Varanasi in collaboration with SEBI and NSDL

1. Problem Statement: Retail Investor Vulnerability

Unprecedented surge in retail market participation meets organized cyber deception:

- Over 150 Million retail Demat accounts in India with exponential growth across Tier-2/Tier-3 cities.
- Proliferation of unregulated 'VIP' Telegram channels, WhatsApp stock tip syndicates, and fake IPO allotment links.
- Deceptive AI deepfakes and impersonation videos targeting unsuspecting retail investors with 300% profit promises.
- Critical lack of accessible, native-language verification tools tailored for non-technical or elderly citizens.

2. Target Users & Core Challenges

Empowering those most exposed to financial misinformation:

- First-Time Retail Investors: Vulnerable to FOMO, false guarantees, and operator penny stock pumps.
- Elderly & Senior Citizens: Frequently targeted by high-pressure phishing messages and account freeze threats.
- Regional Language Users: Millions who communicate in Hindi, Bengali, or regional dialects with zero access to English regulatory portals.
- Voice-Note Consumers: Users who receive financial advice via WhatsApp voice clips rather than formal prospectuses.

3. Proposed Solution: SACH AI

An accessible, voice-first multilingual misinformation & content literacy shield:

- Voice-First Ingestion: Record via browser mic, upload audio notes (.wav/.mp3), or paste text directly.
- Multilingual Support: Full UI and speech pipeline in English, Hindi (■■■■■■■), and Bengali (■■■■■).
- Hybrid AI & NLP Rule Engine: Real-time identification of prohibited guaranteed returns and dabba trading.
- Spoken Plain-Language Explanations: Translates complex regulatory principles into clear audio guidance.
- Evidence-Aware Linking: Directly cross-references official SEBI SCORES, NSDL, BSE/NSE, and 1930 Helpline.

4. System Architecture & Workflow Pipeline

End-to-end processing from spoken audio to verified evidence:

- 1. Audio & Text Capture: Ingests audio streams with real-time waveform and duration monitoring.
- 2. Speech-to-Text: Robust transcription engine with language detection and editable transcript review.
- 3. NLP Claim Analyzer: Tokenizes sentences, extracts specific claims, tags red flag triggers, and calculates 0-100 risk score.
- 4. Regulatory Evidence Mapping: Cites statutory mechanisms (ASBA for IPOs, SCORES for broker grievances).
- 5. Text-to-Speech Output: Generates base64 voice responses (gTTS) for auditory accessibility.

5. Technology Stack & Implementation

Modern, robust, modular full-stack architecture:

- Frontend: React.js 18 with Vite, Tailwind CSS, Lucide Icons, responsive mobile-first UI.
- Backend: Python 3.12, FastAPI asynchronous framework, Pydantic validation, Uvicorn server.
- AI & Speech: SpeechRecognition, gTTS, modular NLP rules engine, optional OpenAI/Gemini integration.
- Database & Telemetry: SQLite store for non-sensitive aggregated audit logs.
- Deployment: Docker & Docker Compose containerization with environment variable secret management.

6. Accessibility, Privacy & Safety by Design

Engineered for public trust and absolute user safety:

- Accessibility Modes: Built-in Large Text mode (A+) and High-Contrast mode for elderly and low-vision users.
- Zero Persistent Audio Retention: Audio files are processed in-memory / temporary buffers and immediately purged.
- Strict Non-Advisory Guardrail: Programmed never to provide stock recommendations, buy/sell targets, or price tips.
- No False Guarantees: Always emphasizes statutory market risk even on safe informational statements.

7. Regulatory & Official Evidence Integration

Seamless alignment with Indian investor protection institutions:

- SEBI SCORES: Direct links for lodging complaints against fraudulent brokers and unlisted advisors.
- SEBI Saa■thi: Educational guides on mutual funds and securities market fundamentals.
- NSDL Depository Services: Demat credit and Consolidated Account Statement (CAS) verification.
- National Cyber Crime Portal & 1930 Helpline: 4-step emergency action plan to freeze defrauded funds in transit.

8. Expected Impact & Scalability

Protecting India's next 100 million retail investors:

- Democratizing Investor Literacy: Bringing regulatory wisdom to regional language speakers across Bharat.
- Proactive Fraud Prevention: Enabling instant 10-second verification before transferring funds to fraudsters.
- Extensible Microservice: Designed for plug-and-play integration as a broker API or WhatsApp Bot.

9. Roadmap & Conclusion

The path forward for investor protection technology:

- Phase 1 (Completed Prototype): Multilingual web app with voice I/O, scam quiz, and evidence mapping.
- Phase 2: Official WhatsApp and Telegram Bot integration for inline forward fact-checking.
- Phase 3: Multimodal video deepfake detection and screenshot OCR analyzer.
- Tagline: "Suniye, Samajhiye, Sachai pe Bharosa Kijiye."